

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1511
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 15
Duration : 60 Minutes

Total Marks: 30

Code of Conduct

I **_Lavanya.M(192321051)_** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **_Lavanya.M_**

MCQ Test

1. A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?

- a) 5 TB
- b) 10 TB
- c) 15 TB
- d) 20 TB

Ans: c) 15 TB

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

- ~4 minutes
- b) ~40 minutes
- c) ~4 hours
- d) ~24 hours

Ans: a) ~4 minutes

3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

- a) 5000
- b) 6000
- c) 7000
- d) 10,000

Ans: b) 6000

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

- 1 hour
- b) 10 hours
- c) 1.1 hours
- d) 5.5 hours

Ans: c) 1.1 hours

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

- a) ~4 hours
- b) ~8 hours
- c) ~20 hours
- d) ~40 hours

Ans: a) ~4 hours

6. A cloud provider offers pay-as-you-go pricing at \$0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

- a) \$50
- b) \$500
- c) \$1000
- d) \$2000

Ans: b) \$500

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

- a) 9.8 Gbps
- b) 9.6 Gbps
- c) 8.0 Gbps
- d) 7.5 Gbps

Ans: a) 9.8 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

- a) 91.25 GB
- b) 85 GB
- c) 100 GB
- d) 120 GB

Ans: a) 91.25 GB

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

- a) 500 Mbps
- b) 250 Mbps
- c) 200 Mbps
- d) 1000 Mbps

Ans: a) 500 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

- a) 262 minutes
- b) 365 minutes
- c) 525 minutes
- d) 150 minutes

Ans: a) 262 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

- a) 4,000
- b) 8,000
- c) 6,000
- d) 10,000

Ans: b) 8000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?

- a. 4 GB
- b) 6 GB
- c) -8 GB
- d) 10 GB

Ans: a) 4GB

13. A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?

- a. 16
- b) 32
- c) 64
- d) 128

Ans: c) 64

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?

- a. 40 kW
- b) 50 kW
- c) 60 kW
- d) 70 kW

Ans: a) 40 kW

15. A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

- a. 6,400
- b) 12,800
- c) 25,600
- d) 50,000

Ans: c) 25,600

16. Which cloud deployment model is used for organizations that require both private and public cloud resources?

- A): Hybrid Cloud
- B): Public Cloud
- C): Private Cloud
- D): Community Cloud

Ans: a) Hybrid cloud

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

- A): Encryption
- B): Load balancing
- C): Virtualization
- D): Resource pooling

Ans: a) Encryption

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

- a: 5000 GB**
- b: 7000 GB**
- c: 6000 GB**
- d: 7500 GB**

Ans: b) 7000 GB

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

- a: 56 GHz**
- b: 58 GHz**
- c: 64 GHz**
- d: 70 GHz**

Ans: a) 56 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

- a: 1.5 sec**
- b: 2 sec**
- c: 1 sec**
- d: 3 sec**

Ans: a) 1.5 sec

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

- a: 600 GB**
- b: 500 GB**
- c: 700 GB**
- d: 800 GB**

Ans: a) 600 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

- a: 21.6 min**
- b: 36 min**
- c: 43.2 min**
- d: 18 min**

Ans: a) 21.6 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

- a: 120 MB**
- b: 100 MB**

c: 150 MB

d: 200 MB

Ans: a) 120

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.

a: 180

b: 150

c: 120

d: 160

Ans: a) 180

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

a: 1200

b: 1000

c: 800

d: 600

Ans: a) 1200

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

a: 12 min

b: 10 min

c: 6 min

d: 8 min

Ans: a) 12 min

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.

Formula:

$\text{Energy} = \text{Units} \times \text{Power} \times \text{Time (hrs)}$

a: 420 kWh

b: 480 kWh

c: 400 kWh

d: 500 kWh

Ans: a) 420 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

- a: 32 TB**
- b: 35 TB**
- c: 30 TB**
- d: 28 TB**

Ans: a) 32 TB

29. Which of the following is NOT a benefit of virtualization?

- i. Reduced hardware costs
- b) Increased energy consumption
- c) Improved resource management
- d) Faster deployment of applications

Ans: b) Increased energy consumption

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

- i. Hypervisor
- b) Load balancer
- c) Firewall
- d) BIOS

Ans: a) Hypervisor